01.03 · THEORY + PAST-PAPER INDEX

Thermodynamics

Temperature, energy and equilibrium in astronomical systems

START HERE

Thermodynamics describes the macroscopic state of matter and how energy is stored and transferred. Starting with temperature, pressure and the ideal gas, we build the first law, common gas processes, radiative balance and the three ways energy moves through an astrophysical environment.

LEARNING GOALS

✓	Distinguish temperature, heat, internal energy and work.
✓	Use the ideal-gas law and first law with a consistent sign convention.
✓	Analyze isothermal, isochoric, isobaric and adiabatic changes.
✓	Set up radiative equilibrium and compare energy-transport mechanisms.

1 · Thermodynamic states and equilibrium

A thermodynamic system contains many particles, but it can often be described by a few measurable state variables. For a gas these include pressure P, volume V, temperature T and amount of matter. Pressure is force per unit area; temperature measures the thermal state and determines the direction in which energy flows when two systems are brought into contact.

THERMAL EQUILIBRIUM

Two systems in thermal equilibrium have no net heat flow between them when placed in contact. Temperature is the property that is equal at equilibrium. Heat is not a substance stored inside an object; it is energy crossing a boundary because of a temperature difference.

An equation of state connects state variables. The ideal-gas model assumes particles are small compared with their separations, move randomly, and interact mainly through brief elastic collisions. It is a good approximation for a dilute gas when the mean particle spacing is large compared with the interaction scale.

$$PV = Nk_BT = nRT$$

N is the number of particles, k_B is Boltzmann's constant, n is the amount in moles, and $R = 8.314\ \text{J mol}^{-1}\ \text{K}^{-1}$. Use kelvins, not degrees Celsius, in gas-law ratios. If N is fixed, doubling the absolute temperature at fixed volume doubles pressure.

2 · Internal energy, work and the first law

Internal energy U is microscopic energy stored in random motion and interactions of a system's particles. For a monatomic ideal gas, the internal energy is entirely translational kinetic energy and depends only on temperature. Heat Q and work W are two ways energy can cross the system boundary.

$$U = (3/2)Nk_BT = (3/2)nRT \text{ (monatomic ideal gas)}$$

SIGN CONVENTION USED HERE

Q is positive when energy enters the gas as heat. W is positive when the gas does work on its surroundings. With this convention the first law is $\Delta U = Q - W$, or $Q = \Delta U + W$. Some textbooks choose work done on the gas as positive; always state the convention before substituting signs.

For a quasi-static pressure change, work done by the gas is $W = \int P\,dV$: the area under a pressure–volume curve. Expansion has positive W; compression has negative W. At constant pressure, $W = P\Delta V$. At fixed volume, no boundary work is done.

$$\Delta U = Q - W \cdot W = \int P\,dV$$

For an ideal gas, a change in internal energy depends on the temperature change, not on the path. The heat needed to cause that change does depend on the path because different processes do different amounts of work.

3 · The four standard gas processes

An isochoric process keeps volume constant: $W = 0$, so all added heat changes internal energy. An isobaric process keeps pressure constant: the gas can expand and do work $P\Delta V$. An isothermal process keeps temperature constant; for an ideal gas $\Delta U = 0$, so heat entering the gas equals the work it performs.

$$\text{Isochoric: } V \text{ constant, } W = 0 \cdot \text{Isobaric: } P \text{ constant, } W = P\Delta V$$

$$\text{Isothermal ideal gas: } T \text{ constant, } \Delta U = 0, Q = W$$

An adiabatic process exchanges no heat with the surroundings, $Q = 0$. A reversible adiabatic change of an ideal gas satisfies $PV^\gamma = \text{constant}$, with $\gamma = C_P/C_V$. For a monatomic ideal gas $\gamma = 5/3$. Expansion cools the gas because it does work while no heat is supplied.

MOLECULAR HEAT CAPACITY

For a monatomic ideal gas $C_V = 3R/2$ and $C_P = 5R/2$ per mole, so $C_P - C_V = R$. The extra heat at constant pressure is needed partly to raise internal energy and partly to do expansion work. More complex molecules store energy in rotational and vibrational modes as well.

A good calculation method is to identify the process first, write what stays fixed, then apply the equation of state and first law. Do not assume $Q = \Delta U$: that is only true when no work is done.

4 · Radiative equilibrium: an astronomical energy balance

A planet intercepts starlight over its projected disk, area πR^2, but emits thermal radiation over its entire spherical surface, area $4\pi R^2$. Let the incident stellar flux at the planet be F and the Bond albedo be A, the fraction of incoming power reflected back to space. In a simple uniform-temperature model, absorbed and emitted powers balance.

$$(1 - A)\, F\, \pi R^2 = 4\pi R^2\, \sigma T^4 \Rightarrow T = [(1 - A)F/(4\sigma)]^{(1/4)}$$

The radius cancels. The factor of four comes from comparing the illuminated disk to the full sphere; it is not an arbitrary correction. If the planet radiates only on its day side, has strong temperature contrasts, an atmosphere, internal heat or wavelength-dependent emissivity, this simple effective temperature is not the local surface temperature.

$$\text{For a star: } F = L/(4\pi d^2) \cdot \text{For a blackbody surface: } F_{surface} = \sigma T^4$$

GREENHOUSE WARMING

An atmosphere can absorb outgoing infrared radiation and emit it both upward and downward. This changes the relation between the planet's effective radiating temperature and its surface temperature. The energy balance still applies to the whole planet at the top of the atmosphere, but the surface need not behave as a bare blackbody.

5 · How energy moves: conduction, convection and radiation

Conduction transfers energy through microscopic interactions between neighboring particles. It is important in solids and across thin interfaces. In a fluid, convection transports energy when warmer, often less-dense material moves and cooler material replaces it. The rising parcel can continue upward if it remains buoyant compared with its surroundings.

Radiation transfers energy through photons and does not require matter in between. A temperature gradient drives net radiative flow because hotter regions emit more energy and the radiation field is not balanced in every direction. In an optically thick stellar interior, photons are repeatedly absorbed and re-emitted; near a transparent surface, radiation can escape.

OPTICAL DEPTH

Optical depth is a measure of how difficult it is for radiation to pass through a medium. Small optical depth means the medium is mostly transparent; large optical depth means photons are likely to interact many times. The photosphere is the layer from which most of the visible light we observe escapes.

In many stars, radiation transports energy through some layers and convection through others. Which mechanism dominates depends on opacity, temperature gradient, density and the way energy is produced. A complete stellar model uses local balance laws; qualitative problems often ask only which transport mechanism is plausible and why.

6 · Thermal time scales and model checks

A time scale compares an energy reservoir with the rate at which energy is lost or supplied. If a body contains usable thermal or gravitational energy E and radiates with luminosity L, a characteristic time is $t \approx E/L$. For a star, the Kelvin–Helmholtz estimate compares gravitational binding energy with luminosity and is of order $GM^2/(RL)$.

This is not a prediction that a real star must contract uniformly: nuclear energy can replenish the radiated energy, and the star can settle into hydrostatic and thermal equilibrium. The estimate is a useful comparison, not a replacement for stellar structure equations.

CHECKLIST

Keep temperature in kelvins, distinguish flux from total luminosity, and track the surface area over which power is absorbed or emitted. On a P–V diagram, expansion work is the signed area under the path. Check energy conservation with $\Delta U = Q - W$ before judging whether a result is physically plausible.

PAST IOAA PROBLEM REFERENCES

The following questions are from official past IOAA problem sets.

Use each year, round and question label to locate the original problem in its official paper.

01	IOAA 2007 · TH 1.9
02	IOAA 2007 · TH Q2
03	IOAA 2008 · TH S4
04	IOAA 2008 · TH S9
05	IOAA 2010 · TH S5
06	IOAA 2010 · TH S12
07	IOAA 2012 · TH S7
08	IOAA 2013 · TH S1
09	IOAA 2013 · TH S3
10	IOAA 2014 · TH P7
11	IOAA 2014 · TH P8
12	IOAA 2014 · TH P10
13	IOAA 2014 · DA D2
14	IOAA 2015 · TH L3
15	IOAA 2015 · TH S9
16	IOAA 2016 · TH T2
17	IOAA 2019 · TH 5
18	IOAA 2020 · TH 2
19	IOAA 2021 · TH Q2
20	IOAA 2022 · TH 5

21 IOAA 2022 · TH 11

22 IOAA 2023 · TH Q4